

INTERNSHIP FINAL REPORT

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Department: CSD / Data Science

Internship Duration: January 27, 2026 – February 28, 2026

Company: Shadow Fox

Domain: Data Science

1. Objectives

The primary objectives of this internship at ShadowFox were:

1. To gain practical exposure to real-world Data Science applications.
2. To understand the complete Data Science lifecycle from data collection to model evaluation.
3. To develop hands-on experience in data preprocessing, visualization, and machine learning.
4. To enhance analytical and problem-solving skills using Python-based tools.

2. Tasks and Responsibilities:

During my internship at Shadow Fox, I worked on the following tasks:

◆ Data Collection and Understanding:

Collected and analyzed structured datasets from CSV files and online repositories. Understood dataset structure, features, and target variables.

◆ Data Cleaning and Preprocessing:

- Handled missing values using mean/median imputation.
- Removed duplicate entries.
- Encoded categorical variables using Label Encoding and One-Hot Encoding.

- Performed feature scaling using StandardScaler.

◇ **Exploratory Data Analysis (EDA):**

- Used Pandas and NumPy for data manipulation.
- Created visualizations using Matplotlib and Seaborn.
- Identified correlations between features using heatmaps.
- Analyzed trends and patterns in the dataset.

◇ **Feature Engineering:**

- Created new features from existing columns.
- Transformed categorical features into numerical format.
- Selected important features for improving model performance.

◇ **Machine Learning Model Implementation:**

Implemented various machine learning algorithms including:

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest

Evaluated models using:

- Accuracy
- Precision
- Recall

- F1-Score

◇ **Model Evaluation and Validation:**

- Performed train-test split.
- Applied cross-validation techniques.
- Compared different models to select the best-performing one.

3.Tools and Technologies Used:

Tools and Technologies Used

- **Python**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Seaborn**
- **Scikit-learn**
- **Jupyter Notebook / Google Colab**

4. Learning Outcomes:

Through this internship, I:

- **Gained strong practical knowledge of Data Science concepts.**
- **Understood the importance of data preprocessing in improving model performance.**
- **Developed skills in data visualization and interpretation.**
- **Learned how to build, evaluate, and optimize machine learning models.**
- **Improved my communication and documentation skills.**

5. Challenges and Solutions:

◇ Handling Missing and Inconsistent Data

I resolved this by applying data cleaning techniques and proper preprocessing methods.

◇ Improving Model Accuracy

I performed feature selection and hyperparameter tuning to enhance performance.

◇ Working with Large Datasets

Optimized data processing using efficient Pandas operations.

6.Conclusion:

The internship at Shadow Fox provided valuable exposure to real-world Data Science practices. It strengthened my understanding of data preprocessing, exploratory data analysis, and machine learning model development. This experience has significantly improved my technical knowledge and motivated me to pursue a professional career in Data Science.

7. Acknowledgment:

I sincerely thank Shadow Fox for providing me with this valuable internship opportunity. I am grateful to my mentor and team members for their guidance and support throughout the internship. I also thank Pallavi Engineering College for encouraging practical learning experiences..